Labs by Carawin — Backend API Routes

Base URL: http://localhost:4000 (configurable via PORT) **API base path:** /api/v1

Authentication legend: ■ requires authentication using a Bearer JWT. Role- and permission-scoped access is shown in the Auth / Role column. The health endpoint is outside the API base path.

1. Health

Method	Path	Notes
GET	/health	DB ping

2. Auth — /api/v1/auth

Method	Path	Auth
POST	/register	public
POST	/login	public
POST	/google	public
POST	/google/link	■
POST	/refresh	public
POST	/logout	public
POST	/logout-all	■
POST	/verify-email	public
POST	/forgot-password	public
POST	/reset-password	public
GET	/me	■
GET	/sessions	■
DELETE	/sessions/:id	■

3. Schools — /api/v1/schools

Method	Path	Auth / Role
POST	/	■ SCHOOL_ADMIN
GET	/	■ SUPER_ADMIN
POST	/enroll	■
GET	/:id	■ school member / SA
GET	/:id/lab-codes	■ school:manage
PUT	/:id/labs	■ school:manage
POST	/:id/verify	■ SUPER_ADMIN
POST	/:id/referral-codes	■ school:manage
GET	/:id/referral-codes	■ school:manage
POST	/:id/referral-codes/:codeId/revoke	■ school:manage
GET	/:id/seats	■ school:manage
GET	/:id/students	■ progress:view
POST	/:id/students/bulk-import	■ students:manage
DELETE	/:id/members/:userId	■ students:manage
POST	/:id/teachers	■ school:manage
GET	/:id/teachers	■ school:manage
POST	/:id/classes	■ classes:manage
GET	/:id/classes	■ SCHOOL_ADMIN / TEACHER
POST	/:id/classes/:classId/members	■ classes:manage
GET	/:id/labs	■ school:manage

4. Labs — /api/v1/labs

Method	Path	Auth / Role
POST	/	■ labs:create
GET	/	■
GET	/:id	■
PUT	/:id	■ labs:create
DELETE	/:id	■ labs:create
POST	/:id/modules	■ labs:create
PUT	/:id/modules/:moduleId	■ labs:create
DELETE	/:id/modules/:moduleId	■ labs:create
PUT	/:id/assessment	■ labs:create
PUT	/:id/experiment	■ labs:create
PUT	/:id/badge	■ labs:create
POST	/:id/submit-review	■ labs:create
POST	/:id/approve	■ SUPER_ADMIN
POST	/:id/reject	■ SUPER_ADMIN
POST	/:id/publish	■ labs:create
POST	/:id/archive	■ labs:create
PUT	/:id/status	■ labs:create
POST	/:id/trial	■ SUPER_ADMIN
GET	/:id/materials	■ (entitlement)
GET	/:id/experiment	■ (entitlement)

4a. Simulation upload — /api/v1/labs/:id/simulation

Method	Path	Auth / Role
POST	/upload-urls	■ labs:create
POST	/commit	■ labs:create
GET	/	■ labs:create
DELETE	/	■ labs:create
GET	/file	■ labs:create
PUT	/file	■ labs:create

5. Uploads — /api/v1/uploads

Method	Path	Auth
POST	/presign	■

6. Simulation player — /api/v1/sim

Method	Path	Auth
GET	/:token/*path	signed sim token in URL

7a. Razorpay — /api/v1/payments

Method	Path	Auth
POST	/webhook	HMAC signature (mounted with express.raw)
POST	/create-order	■ plans:purchase
POST	/dev/simulate/:orderId	■ (dev only)
GET	/history	■

7b. PayU — /api/v1/payments/payu

Method	Path	Auth
POST	/initiate	■
POST	/callback	PayU signature

8. Assessments — /api/v1/assessments

Method	Path	Auth
POST	/start	■
POST	/attempts/:id/submit	■
GET	/attempts/:id/result	■

9. Teacher verification — /api/v1/teacher

Method	Path	Auth
GET	/verifications	■ practical:verify
GET	/verifications/:id	■ practical:verify
POST	/verifications/:id/approve	■ practical:verify
POST	/verifications/:id/reject	■ practical:verify
GET	/students	■ practical:verify
GET	/reports	■ practical:verify

10. Notifications — /api/v1/notifications

Method	Path	Auth
GET	/	■
POST	/:id/read	■
POST	/read-all	■

11. Admin — /api/v1/admin

Method	Path	Auth / Role
GET	/dashboard	SUPER_ADMIN (router-level guard)
GET	/users	SUPER_ADMIN (router-level guard)
POST	/users	SUPER_ADMIN (router-level guard)
PATCH	/users/:id/status	SUPER_ADMIN (router-level guard)
GET	/payments	SUPER_ADMIN (router-level guard)
GET	/subscriptions	SUPER_ADMIN (router-level guard)
GET	/audit-logs	SUPER_ADMIN (router-level guard)

12a. Public badges — /api/v1/badges

Method	Path	Auth
GET	/:id/verify	unauthenticated
GET	/:id/qr	unauthenticated

12b. Authenticated badges — /api/v1/badges

Method	Path	Auth / Role
GET	/	■
GET	/mine	■
GET	/:id	■ (owner / SA)
POST	/:id/revoke	■ SUPER_ADMIN

12c. Current user's badges — /api/v1/users

Method	Path	Auth
GET	/me/badges	■

13. Plans & Subscriptions — /api/v1

Method	Path	Auth / Role
GET	/plans	■
POST	/plans	■ SUPER_ADMIN
PUT	/plans/:id	■ SUPER_ADMIN
DELETE	/plans/:id	■ SUPER_ADMIN
POST	/subscriptions	■ plans:purchase
GET	/subscriptions/current	■
PATCH	/subscriptions/:id/status	■ SUPER_ADMIN

14. Learning — /api/v1

Method	Path	Auth
POST	/labs/:id/start	■ (student/individual)
GET	/labs/:id/progress	■
POST	/lessons/:id/progress	■
POST	/lessons/:id/complete	■
POST	/experiments/sessions	■
POST	/experiments/sessions/:id/complete	■
POST	/attempts/:id/verification	■ (school student)